MB403 WEEK 5 2020/21

REDUCTION OF ORDER for second order homogeneous equations

Consider the second order homogeneous equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0 \quad — (1)$$

Let y_1 be a nontrivial solution to (1) on an interval I. The second linearly independent solution of (1) is $y_2 = y_1 v(x)$, where $v(x)$ can be determined by substituting y_2 into equation (1). This substitution reduces the equation to a first-order variable separable equation in v'. The general solution is $y = C_1 y_1 + C_2 y_2$

Examples

y_1 is the solution to the given equation. Use reduction of order to find the general solution

(1) $y'' - 4y' + 4y = 0$; $y_1 = e^{2x}$

(2) $y'' - y = 0$; $y_1 = \cosh x$

Soln

Let $y_2 = ve^{2x}$

$\Rightarrow y_2' = v'e^{2x} + 2ve^{2x}$

$\Rightarrow y_2'' = v''e^{2x} + 4e^{2x}v' + 4ve^{2x}$

Substituting

$v''e^{2x} + 4v'e^{2x} + 4ve^{2x} - 4v'e^{2x} - 8ve^{2x} + 4ve^{2x} = 0$

$\Rightarrow v''e^{2x} = 0$

$\Rightarrow u'e^{2x} = 0$ where $u = v'$

$\Rightarrow u' = 0$

$\Rightarrow u = K_1 = v'$

$\Rightarrow v = K_1 x + K_2$

$y_2 = k_1 x e^{2x} + k_2 e^{2x}$

$y = c_1 y_1 + c_2 y_2$

$= c_1 e^{2x} + c_2 k_1 x e^{2x} + c_2 k_2 e^{2x}$

$= A e^{2x} + B x e^{2x}$ //

(2). Let $y_2 = v \cosh x$

$\Rightarrow y_2' = v' \cosh x + v \sinh x$

$\Rightarrow y_2'' = v'' \cosh x + 2v' \sinh x + v \cosh x$

substituting

$v'' \cosh x + 2v' \sinh x + v \cosh x - v \cosh x = 0$

$\Rightarrow v'' \cosh x + 2v' \sinh x = 0$

$\Rightarrow u' \cosh x + 2u \sinh x = 0$ where $u = v'$

$\Rightarrow u' + 2u \tanh x = 0$

$\Rightarrow \frac{u'}{u} = -2 \tanh x$

$\Rightarrow \ln u = -2 \ln(\cosh x) + \ln C$

$\Rightarrow \ln(Cu) = \ln(\operatorname{sech}^2 x)$

$\Rightarrow Cu = \operatorname{sech}^2 x$

$\Rightarrow u = \frac{1}{C} \operatorname{sech}^2 x = v'$

$\Rightarrow v = C_1 \tanh(x) + C_2$

$y_2 = C_1 \sinh x + C_2 \cosh x$

$\therefore y = A \cosh x + B \sinh(x)$

Reduction Of Order for second order nonhomogeneous equations

Consider the second order eqⁿ

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = f(x),\ f(x) \neq 0 \quad \text{——} (1)$$

Let y_1, defined on interval I, be a nontrivial solution to the homogeneous eqⁿ associated to (1).
Substituting $y_2 = y_1 v$ into (1) reduces (1) to a first-order linear equation in v'. The general solution to (1) is $y = y_2$

Examples
y_1 is the solⁿ to the associated homogeneous eqⁿ.
Find the general solⁿ of the given eqⁿ

(1) $y'' - 4y = 2$; $y_1 = e^{-2x}$

(2) $y'' - 3y' + 2y = 5e^{3x}$; $y_1 = e^{x}$

solⁿ

Let $y_2 = ve^{-2x}$

$\Rightarrow y_2' = v'e^{-2x} - 2e^{-2x}v$

$\Rightarrow y_2'' = v''e^{-2x} - 4v'e^{-2x} + 4ve^{-2x}$

Substituting

$v''e^{-2x} - 4v'e^{-2x} + 4ve^{-2x} - 4ve^{-2x} = 2$

$\Rightarrow v'' - 4v' = 2e^{2x}$

$\Rightarrow u' - 4u = 2e^{2x}$, $u = v'$

Integrating factor is

$u = e^{\int -4dx}$

$= e^{-4x}$

Multiplying we get

$e^{-4x}u' - 4e^{-4x}u = 2e^{-2x}$

$$\Rightarrow \frac{d}{dx}(ue^{-4x}) = 2e^{-2x}$$

$$\Rightarrow ue^{-4x} = -2e^{-2x} + C_1$$

$$\Rightarrow u = -e^{2x} + C_1 e^{4x} = v'$$

$$\Rightarrow v = -\frac{1}{2}e^{2x} + \frac{C_1}{4}e^{4x} + C_2$$

$$\therefore y = Ae^{-2x} + Be^{2x} - \frac{1}{2}$$

(2) Let $y_2 = ve^x$

$$\Rightarrow y_2' = v'e^x + ve^x$$

$$\Rightarrow y_2'' = v''e^x + 2v'e^x + ve^x$$

Substituting

$$v''e^x + 2v'e^x + ve^x - 3v'e^x - 3ve^x + 2ve^x = 5e^{3x}$$

$$\Rightarrow v''e^x - v'e^x = 5e^{3x}$$

$$\Rightarrow u' - u = 5e^{2x}, \quad u = v'$$

Integrating factor

$$u = e^{\int -dx}$$

$$= e^{-x}$$

Multiplying we get

$$e^{-x}u' - e^{-x}u = 5e^x$$

$$\Rightarrow \frac{d}{dx}(ue^{-x}) = 5e^x$$

$$\Rightarrow ue^{-x} = 5e^x + C_1$$

$$\Rightarrow u = 5e^{2x} + C_1 e^x = v'$$

$$\Rightarrow v = \frac{5}{2}e^{2x} + C_1 e^x + C_2$$

$$\therefore y = Ae^x + Be^{2x} + \frac{5}{2}e^{3x}$$

Reduction of Order in general

Reduction of order can be used to solve any higher order differential eqn, homogeneous or nonhomogeneous. For n^{th} order eqn, if y_1, which is the soln of the associated homogeneous eqn, is known, then the eqn can be reduced to an $(n-1)^{th}$ order eqn in $v^{(n-1)}$. If the soln to the homogeneous eqn associated with the resulting eqn is known, then the process can be repeated until we get the soln.

Inverse Operator Method

Consider a linear differential equation with constant coefficients

$a_n D^n y + a_{n-1} D^{n-1} y + \cdots + a_1 Dy + a_0 y = f(x)$, $f(x) \neq 0$

which can be written as

$$F(D)\, y = f(x) \quad \text{---} \quad (1)$$

where $F(D) = a_n D^n + a_{n-1} D^{n-1} + \cdots + a_1 D + a_0$ is a differential operator.

From (1), we get the particular solution as $y_p = \frac{1}{F(D)} f(x)$ where $\frac{1}{F(D)}$ is an inverse differential operator.

With the help of the following useful formulae, we look at the application this method for specific forms of the nonhomogeneous term $f(x)$.

(1). $F(D)[e^{ax} V(x)] = e^{ax} F(D+a) V(x)$ This is known as the shift formula

(2). $\frac{1}{F(D)} e^{ax} = \frac{e^{ax}}{F(a)}$ if $F(a) \neq 0$

If $F(a) = 0$, then $F(D) = (D-a)^n g(D)$, $g(D) \neq 0$

so,

$$\frac{1}{F(D)} e^{ax} = \frac{1}{(D-a)^n g(D)} e^{ax}$$

$$= \frac{1}{g(a)} \left[\frac{1}{(D-a)^n} e^{ax} \cdot 1 \right]$$

$$= \frac{e^{ax}}{g(a)} \frac{1}{D^n}(1) \quad \text{using shift formula}$$

$$= \frac{e^{ax}}{g(a)} \frac{x^n}{n!}$$

3. Recall Euler's formula

$$e^{i(ax+b)} = \cos(ax+b) + i\sin(ax+b)$$

so

$$\frac{1}{F(\delta)}\cos(ax+b) = \mathrm{Re}\left[\frac{1}{F(\delta)}e^{i(ax+b)}\right] \text{ and}$$

$$\frac{i}{F(\delta)}\sin(ax+b) = \mathrm{Im}\left[\frac{1}{F(\delta)}e^{i(ax+b)}\right]$$

Where "Re" is real part of, and "Im" is imaginary part of

4. $\frac{1}{F(\delta)}P(x) = \frac{1}{1+g(\delta)}P(x)$ where $P(x)$ is a polynomial

Then expand $\frac{1}{1+g(\delta)}$ by binomial expansion

Binomial Series; for any real number K

$$(1+x)^K = \sum_{n=0}^{\infty}\binom{K}{n}x^n = 1 + Kx + \frac{K(K-1)}{2!}x^2 + \frac{K(K-1)(K-2)}{3!}x^3 + \cdots$$

Alternative to binomial expansion formula, you can use long division to work out

$$1+g(\delta)\overline{)\,P(x)}$$

Note that unlike in normal long division, you start division with 1 (a constant term of the divisor), instead of the term with the highest degree of δ.

Examples

Find a particular solution to

① $(D^2-4D+3)y = e^{2x}$

② $(D^3+1)y = 3+e^{-x}+5e^{2x}$

③ $(D^4+10D^2+9)y = \cos(2x+3)$

④ $(D^4-1)y = \sin x$

⑤ $(D^2-4)y = \cosh(2x-1)+3^x$

⑥ $(D^3+D^2+1)y = x^2+2x+1$

⑦ $(D^3+D^2+D+1)y = x^4+2x+1$

Sol^n –

① $(D^2-4D+3)y = e^{2x}$

$$\Rightarrow y_p = \frac{1}{D^2-4D+3}\, e^{2x}$$

$$= \frac{e^{2x}}{4-8+3}$$

$$= -e^{2x}$$

② $(D^3+1)y = 3+e^{-x}+5e^{2x}$

$$\Rightarrow y_p = \frac{1}{D^3+1}\,(3+e^{-x}+5e^{2x})$$

$$= \frac{1}{D^3+1}\,3e^{0x} + \frac{1}{(D+1)(D^2-D+1)}\,e^{-x} + \frac{1}{D^3+1}\,5e^{2x}$$

$$= \frac{3}{0+1} + \frac{e^{-x}}{(-1)^2+1+1}\,\frac{x^1}{1!} + \frac{5e^{2x}}{2^3+1}$$

$$= 3 + \frac{xe^{-x}}{3} + \frac{5}{9}e^{2x}$$

③ $(D^4+10D^2+9)y = \cos(2x+3)$

$$\Rightarrow y_p = \frac{1}{D^4+10D^2+9}\cos(2x+3)$$

$$= \mathrm{Re}\left[\frac{1}{D^4+10D^2+9} e^{i(2x+3)}\right]$$

$$= \mathrm{Re}\left[e^{i3}\frac{1}{D^4+10D^2+9} e^{i2x}\right]$$

$$= \mathrm{Re}\left[e^{i3}\frac{e^{i2x}}{-15}\right]$$

$$= \mathrm{Re}\left[\frac{-e^{i(2x+3)}}{15}\right]$$

$$= \mathrm{Re}\left[-\frac{1}{15}\left(\cos(2x+3)+i\sin(2x+3)\right)\right]$$

$$= -\frac{1}{15}\cos(2x+3)$$

④ $(D^4-1)y = \sin x$

$$\Rightarrow y_p = \frac{1}{D^4-1}\sin x$$

$$= \mathrm{Im}\left[\frac{1}{D^4-1} e^{ix}\right]$$

$$= \mathrm{Im}\left[\frac{1}{(D^2-1)(D^2+1)} e^{ix}\right]$$

$$= \mathrm{Im}\left[-\frac{1}{2}\left[\frac{1}{(D-i)(D+i)} e^{ix}\right]\right]$$

$$= \mathrm{Im}\left[-\frac{1}{4i}\left(\frac{1}{D-i}-\frac{1}{D+i}\right)e^{ix}\right]$$ use partial fractions

$$= \mathrm{Im}\left[-\frac{i}{4i}\left(xe^{ix}-\frac{e^{ix}}{2i}\right)\right]$$

$$= \mathrm{Im}\left[\frac{ix\,e^{ix}}{4}-\frac{1}{8}e^{ix}\right]$$

$$= \frac{x}{4}\cos x - \frac{1}{8}\sin x \;//$$

(5) $(D^2-4)\,y = \cosh(2x-1) + 3^x$

$$\Rightarrow y_p = \frac{1}{D^2-4}\left[\cosh(2x-2) + 3^x\right]$$

$$= \frac{1}{D^2-4}\left[\frac{e^{2x-1}+e^{-(2x-1)}}{2} + e^{x\ln 3}\right]$$

$$= \frac{1}{2}\left[e^{-1}\frac{1}{(D-2)(D+2)}e^{2x} + e\,\frac{1}{(D-2)(D+2)}e^{-2x}\right] + \frac{1}{D^2-4}e^{x\ln 3}$$

$$= \frac{1}{2}\left[\frac{e^{-1}}{4}\frac{1}{D-2}e^{2x} - \frac{e}{4}\frac{1}{D+2}e^{-2x}\right] + \frac{e^{x\ln 3}}{(\ln 3)^2-4}$$

$$= \frac{1}{2}\left[\frac{xe^{2x-1}}{4} - \frac{xe^{-(2x-1)}}{4}\right] + \frac{3^x}{(\ln 3)^2-4}$$

$$= \frac{x}{4}\left(\frac{e^{2x-1}-e^{-(2x-1)}}{2}\right) + \frac{3^x}{(\ln 3)^2-1}$$

$$= \frac{x}{4}\sinh(2x-1) + \frac{3^x}{(\ln 3)^2-1}$$

(6) $(D^3+D^2+1)\,y = x^2+2x+1$

$$\Rightarrow y_p = \frac{1}{1+D^3+D^2}(x^2+2x+1)$$

$$= (1-(D^3+D^2)+\cdots)(x^2+2x+1)$$

$$= x^2+2x+1 - D^3(x^2+2x+1) - D^2(x^2+2x+1) + 0$$

$$= x^2+2x+1 - 0 - (2)$$

$$= x^2+2x-1$$

OR

$$1+D^2+D^3 \;\overline{)\; x^2+2x+1 \;(\; x^2+2x-1}$$

$$-(x^2+2+0)$$

$$2x-1$$

$$-(2x+0)$$

$$-1$$

$$(-1)$$

$$0$$

$\therefore y_p = x^2+2x-1$

(7) $(D^3+D^2+D+1)\,y = x^4+2x+1$

$$y_p = \frac{1}{D^3+D^2+D+1}(x^4+2x+1)$$

$$1+D+D^2+D^3 \;\overline{)\; x^4+2x+1 \;}\;\; x^4-4x^3+2x+23$$

$$-(x^4+4x^3+12x^2+24x)$$

$$-4x^3-12x^2-22x+1$$

$$-(-4x^3-12x^2-24x-24)$$

$$2x+25$$

$$-(2x+2)$$

$$23$$

$$-(23)$$

$$\therefore y_p = x^4-4x^3+2x+23$$